

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS & TNAS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	B	B	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,C	A,C	A,B,C	A,B,C	A,B,C	A,C,D	D	B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	6.00	1.00	8.00	2.00	1.60 to 1.70	1.00		

PART-2 : CHEMISTRY

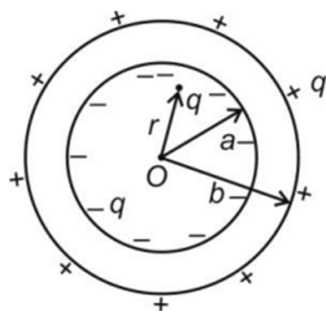
SECTION-I (i)	Q.	1	2	3	4				
	A.	A	D	B	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,D	A,B,D	B,D	A,B,D	A,B,D	B,C	D	B,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	0.60	1605.00	5.00	3.00	78.00	3.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	C	C	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	B,C	A,B	B	A,C,D	A,B,C,D	A,B,C	A,B,C	A,B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	16.00	7.00	2.00	0.11	5.00	4.00		

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A)

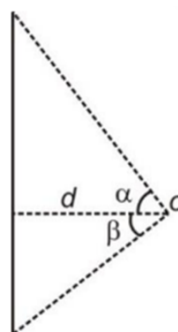

 $Q_{\text{induced, inner}} = -q$ (at distance a from O)

 $Q_{\text{induced, outer}} = +q$ (at distance b from O)

$$\Rightarrow V_0 = \frac{Kq}{b} + \frac{K(-q)}{a} + \frac{Kq}{r}$$

$$= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} - \frac{1}{a} + \frac{1}{b} \right]$$

2. Ans (B)



$$B_0 = \frac{\mu_0 I_0}{4\pi d} [\sin \alpha + \sin \beta]$$

$$B_P = 2 \frac{\mu_0 I_0}{4\pi d} [\sin 53^\circ + \sin 90^\circ]$$

$$= \frac{\mu_0 I_0}{(2\pi)(10^{-2})} \left(\frac{1 + \frac{4}{5}}{\frac{3}{5}} \right)$$

$$= \frac{2(10^{-7})I_0}{(10^{-2})} (3) = 6 \times 10^{-5} T$$

3. **Ans (B)**

1st surface refraction

$$\frac{1.5}{V_1} = \frac{1}{-\left(\frac{R}{N}\right)} = \frac{1.5 - 1}{R} \quad \dots(1)$$

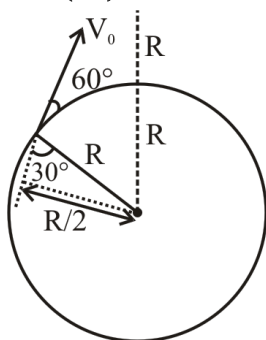
$$\Rightarrow V_1 = \frac{(3R/2N)}{\left(\frac{1}{2N} - 1\right)}$$

Apply refraction from 2nd surface

$$\Rightarrow \frac{1}{\infty} - \frac{1.5}{4} = \frac{1 - 1.5}{-R}$$

On solving we get $N = 2$.

4. **Ans (B)**



From conservation of angular momentum,

$$mv_0 \frac{R}{2} = mv_2 R$$

$$V = \frac{V_0}{4}$$

when gravity is variable during projectile motion, trajectory is not parabolic.

PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (A,C)**

$$dV = \frac{\sigma x d\theta dx}{4\pi\epsilon_0 x}$$

$$\sigma \int_{-\pi/2}^{\pi/2} d\theta \int_0^1 dx$$

$$dV = \frac{\sigma \int_{-\pi/2}^{\pi/2} d\theta \int_0^1 dx}{4\pi\epsilon_0}$$

$$\sigma \int_{-\pi/2}^{\pi/2} 2r \cos \theta d\theta$$

$$dV = \frac{\sigma \int_{-\pi/2}^{\pi/2} 2r \cos \theta d\theta}{4\pi\epsilon_0}$$

$$V = \frac{\sigma r}{\pi\epsilon_0}$$

$$dU = \int_0^R \frac{\sigma r}{\pi\epsilon_0} (\sigma 2\pi r dr)$$

$$U = \frac{2}{3} \frac{\pi\epsilon_0^2 R^3}{\pi\epsilon_0}$$

7. **Ans (A,B,C)**

In the initial state, charge on each capacitor is shown in figure-1.

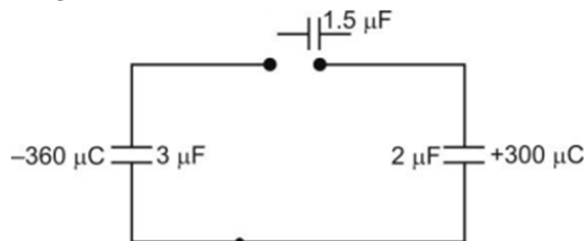


figure - 1

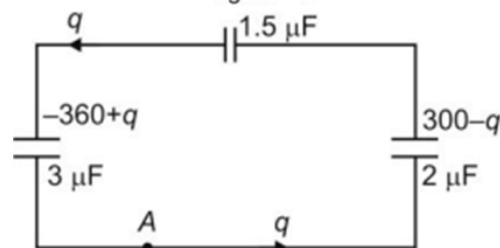


figure - 2

Let charge q flow anti-clockwise in the circuit before it achieves steady state as shown in figure-2.

Applying KVL to figure-2.

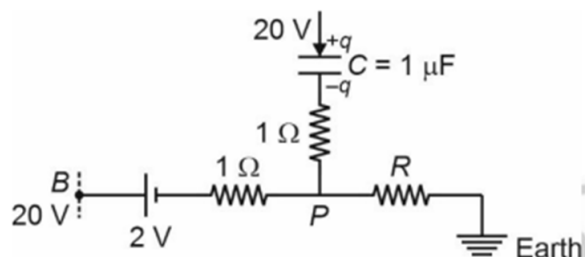
$$-\frac{(360 - q)}{3} + \frac{q}{1.5} - \frac{(300 - q)}{2} = 0$$

$$\Rightarrow q = 180 \mu C$$

$\therefore$ Final charge on $1.5 \mu F$ capacitor is $q = 180 \mu C$ and final charge on $2 \mu F$ capacitor is $300 - q = 120 \mu C$.

8.

Ans (A,B,C)



$$i_{AP} = \frac{dq_3}{dt} = 3e^{-t}$$

At $t = 0$, $q = 0 C$, $i_{AP} = 3 A$ and $\Delta V_C = 0$

$$\therefore 20 - 3 \times 1 = V_P \Rightarrow V_P = 17V$$

Along

$$BP = 20 - 2 \times i \times 1 = V_P = 17 \Rightarrow i = 1A$$

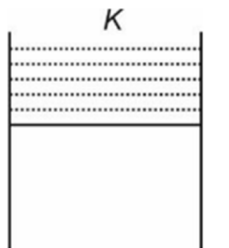
$\therefore$ Current through R is $1 + 3 = 4 A$

$$\text{Value of } R \text{ is } \frac{V_P - 0}{4} = \frac{17}{4} = 4.25 \Omega$$

9. **Ans (A,B,C)**

Using the conservation of charge principle, distribute the charge on all the surfaces of the plates a, b, c, d, e and f. Then use the concept, electric field is zero inside the plates.

11. **Ans (D)**



$$C = \frac{(K+2)\epsilon_0 A}{3d}$$

$$C_2 = \frac{K\epsilon_0 A}{3d}$$

$$\frac{C}{C_2} = \frac{2+K}{K}$$

12. **Ans (B,C)**

$$C_0 = 4\pi\epsilon_0 2R$$

$$C = \frac{2\pi\epsilon_0 \ell}{\ln 2}$$

$$\Rightarrow \ell = 4R \ln 2$$

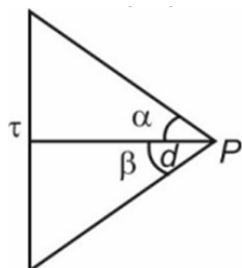
$$8\pi\epsilon_0 R = \frac{\epsilon_0 \times \pi R^2}{d}$$

$$d = \frac{R}{8}$$

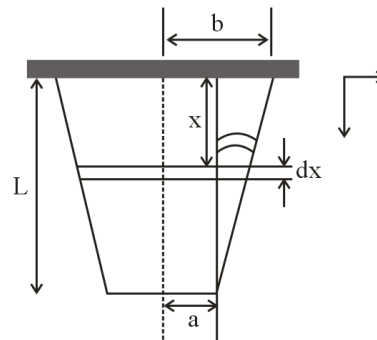
PART-1 : PHYSICS

SECTION-II

1. **Ans (6.00)**



2. **Ans (1.00)**



$$Y = \frac{F/A}{dy/dx}, r = b - \left(\frac{b-a}{L}\right)x$$

$$r = b - \tan \theta x, A = \pi(b - x \tan \theta)^2$$

$$\int_0^{\Delta L} dy = \frac{F}{\pi Y} \int_0^L \frac{dx}{(b - x \tan \theta)^2}$$

$$\Delta L = 10^{-3} \text{ m} = 1 \text{ mm}$$

4. **Ans (2.00)**

Net loss = k (temperature different between both sphere & surrounding)

$$ms \frac{d\theta}{dt} = -k[\theta - \theta_s]$$

$$\int_{\theta_2=0}^{\theta_2=\frac{s^2}{2}} \frac{Cd\theta}{(\theta - \theta_s)} = \int_0^t kdt$$

$$= -C[\ln(\theta - \theta_s)]_0^{\frac{s^2}{2}} = kt$$

$$-C\ln\left(\frac{-\theta_s}{2}\right) = kt$$

$$-C\ln\left(\frac{1}{2}\right) = kt$$

$$C\ln(2) = kt$$

$$t = \frac{C}{k} \ln(2)$$

5. **Ans (1.60 to 1.70)**

$$ms \left(\frac{dT}{dt}\right) = a\sigma AT_0^4 - e\sigma AT_b^4 + 2$$

$$= 2.177 - 0.6889 + 2$$

$$= 3.4831 \text{ watt}$$

$$2 \times 0.25 \times 4.2 \left(\frac{dT}{dt}\right) = 3.4831$$

$$\frac{dT}{dt} = 1.66 \text{ K/sec}$$

$$1.60 \text{ to } 1.70 \text{ K/sec}$$

6. Ans (1.00)

$$v_{PJ} = (i\lambda_{30}) = \left(\frac{10}{R+1} \right) R \text{ for 1st case}$$

$$v_{PJ} = i\lambda_{10} = \left(\frac{5R}{R+2} \right) \text{ for 2nd case}$$

taking ratio

$$3 = \frac{2(R+2)}{(R+1)} \Rightarrow R = 1$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (A)



$$100 - x \quad x \quad x$$

$$100 + x = 140$$

$$x = 40$$

$$A = 60 \text{ mm}$$

$$k = \frac{2.303}{30} \log_{10} \frac{100}{60}$$

$$k = \frac{2.303}{30} \times (1 - 0.8)$$

$$k = \frac{2.303 \times 0.2}{30}$$

$$t_{1/2} = \frac{2.303 \times 0.3}{2.303 \times 0.2} \times 30 = 45 \text{ min}$$

2. Ans (D)

According to RDS method :

$$\text{Rate} = k_2 [\text{N}_2\text{O}_2] [\text{H}_2]$$

$$\text{Where } K_1 = \frac{[\text{N}_2\text{O}_2]}{[\text{NO}]^2}$$

$$[\text{N}_2\text{O}_2] = K_1 [\text{NO}]^2$$

$$\text{So, } r(t) = k_2 K_1 [\text{NO}]^2 [\text{H}_2]$$

$$\text{So, } r(t) = k_2 K_1 [\text{NO}]^2 [\text{H}_2]$$

$$-\frac{1}{2} \frac{d[\text{H}_2]}{dt} = -\frac{1}{2} \frac{d[\text{NO}]}{dt} = \frac{d[\text{N}_2]}{dt}$$

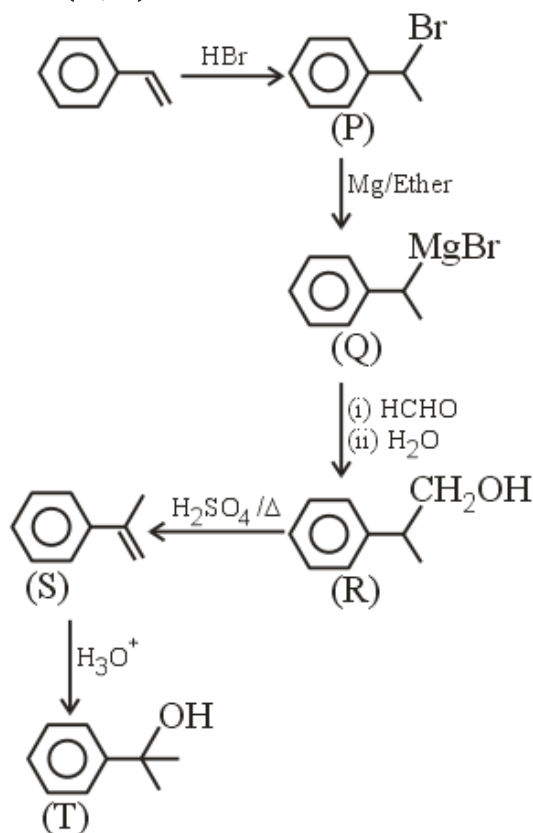
$$= K [\text{NO}]^2 [\text{H}_2]$$

$$\text{Where } K = K_1 k_2$$

PART-2 : CHEMISTRY

SECTION-I (ii)

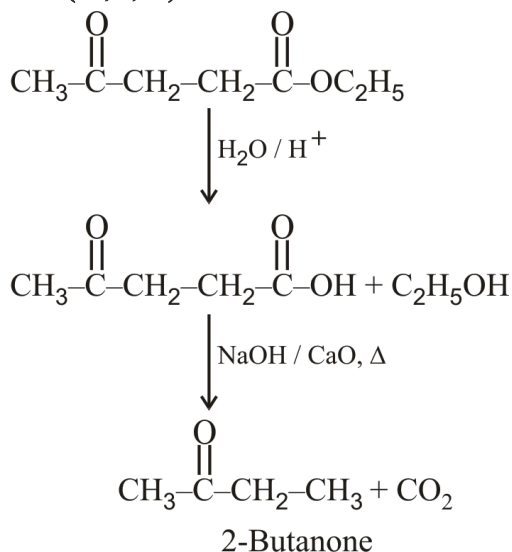
7. Ans (B,D)



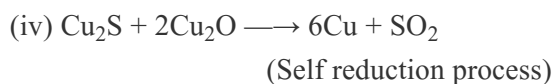
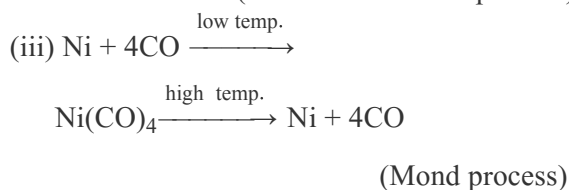
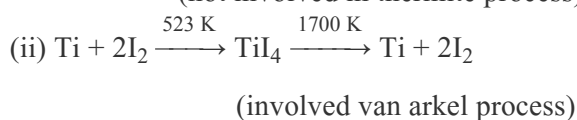
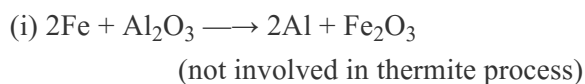
8. Ans (A,B,D)

Comp.	Fehling Test	Tollen's test
HCHO	Y	Y
	N	N
	Y	Y
	N	N
	Y	Y
	N	Y
	N	N

9. Ans (A,B,D)



11. Ans (D)



PART-2 : CHEMISTRY

SECTION-II

1. Ans (0.60)

$$Q + W = \Delta U$$

$$\Rightarrow y - \frac{y}{2} = \Delta U = nC_{v,m} \Delta T$$

$$y = 2n C_{v,m} \Delta T$$

$$\text{using } Q = nC_m \Delta T$$

$$\Rightarrow nC_m \Delta T = 2nC_{v,m} \Delta T$$

$$\Rightarrow C_m = 2C_{v,m} = 2 \times \frac{5}{2}R = 5R$$

$$\text{using } C_m = C_{v,m} + \frac{R}{1-x}$$

$$\Rightarrow \frac{5}{2} = \frac{1}{1-x}$$

$$\Rightarrow x = \frac{3}{5} = 0.60$$

2. Ans (1605.00)

$$T_A = 300$$

$$\frac{T_B}{T_A} = \frac{V_B}{V_A} = \frac{2V}{V}$$

$$T_B = 2T_A = 2 \times 300 = 600$$

$$T_C = 600$$

$$T_D = 75$$

and volume at D will be

$$T_A V_A^{\gamma-1} = T_D V_D^{\gamma-1}$$

$$\left(\frac{300}{75}\right)^{1/\gamma-1} V = V_D, V_D = 8V$$

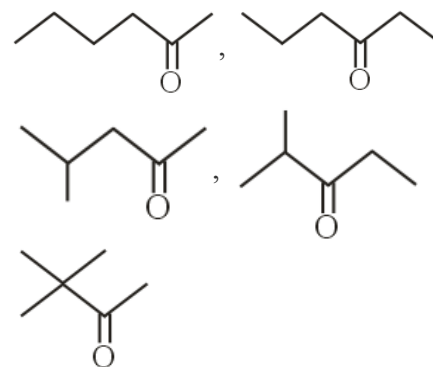
$$W = W_{AB} + W_{BC} + W_{CD} + W_{DA}$$

$$= [-1 \times R \times (600 - 300)] + [-1 \times R \times 600 \ln \frac{8V}{2V}] +$$

$$0 + [1 \times \frac{3}{2} R (300 - 75)]$$

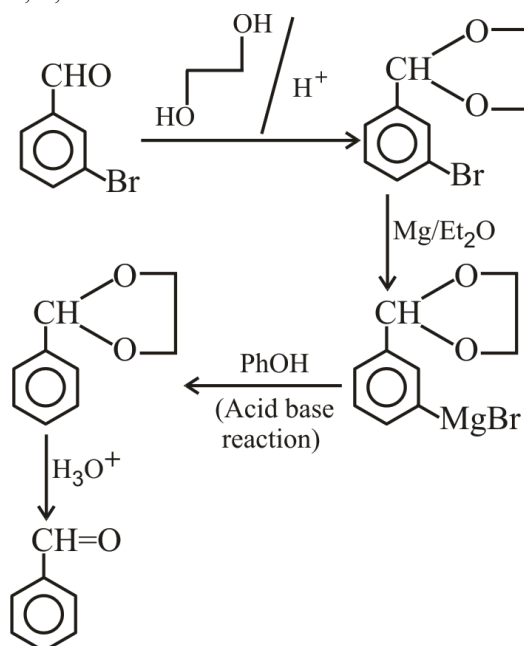
$$= -1605 \text{ cal}$$

3. Ans (5.00)

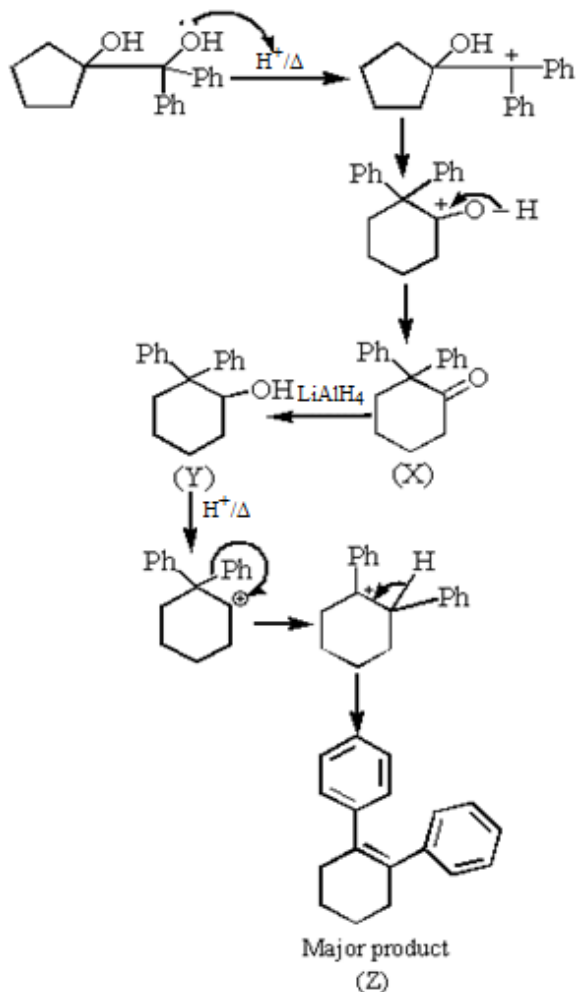


4. Ans (3.00)

a, b, e



5. Ans (78.00)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A)

$$f(x) = \begin{cases} x^2 e^{2(x-1)}; & 0 \leq x \leq 1 \\ a \cos(2x-2) + bx^2; & 1 < x \leq 2 \end{cases}$$

$$\Rightarrow a + b = 1$$

$$f'(x) = \begin{cases} 2xe^{2(x-1)} + 2x^2e^{2(x-1)}; & 0 \leq x \leq 1 \\ 2bx - 2a \sin(2x-2); & 1 < x \leq 2 \end{cases}$$

$$\Rightarrow b = 2$$

2. Ans (C)

We have

$$I = \int_{-\infty}^a \left(\frac{\sin^{-1} e^x + \cos^{-1} e^x}{\tan^{-1} e^a + \tan^{-1} e^x} \right) \left(\frac{e^x}{e^{2x} + 1} \right) dx$$

$$= \frac{\pi}{2} \int_{-\infty}^a \frac{1}{(\tan^{-1} e^a + \tan^{-1} e^x)} \left(\frac{e^x}{(e^{2x} + 1)} \right) dx$$

$$\text{Put } \tan^{-1} e^x = t \Rightarrow \frac{e^x}{e^{2x} + 1} dx = dt$$

$$I = \frac{\pi}{2} \int_0^{\tan^{-1} e^a} \frac{dt}{(t + \tan^{-1} e^a)}$$

$$= \frac{\pi}{2} [\ln(t + \tan^{-1} e^a)]_0^{\tan^{-1} e^a}$$

$$= \frac{\pi}{2} [\ln(2 \tan^{-1} e^a) - \ln(\tan^{-1} e^a)]$$

$$= \frac{\pi}{2} \ln 2$$

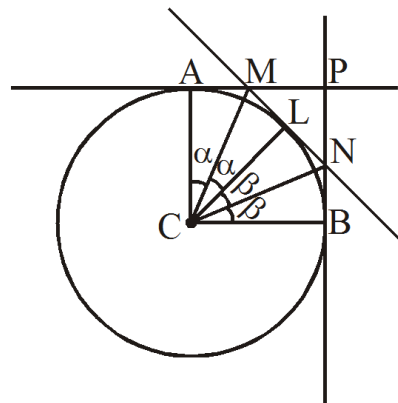
3. Ans (C)

$$y = \sin^{-1}(2x\sqrt{1-x^2}) + \sin^{-1}x$$

$$y = \begin{cases} -\pi - \sin^{-1}x & x \in [-1, -\frac{1}{\sqrt{2}}] \\ 3\sin^{-1}x & x \in [-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}] \\ \pi - \sin^{-1}x & x \in [\frac{1}{\sqrt{2}}, 1] \end{cases}$$

$$\frac{dy}{dx} \text{ can be } \frac{-1}{\sqrt{1-x^2}} \text{ or } \frac{3}{\sqrt{1-x^2}}$$

4. Ans (A)



$$\text{Since, } 2\alpha + 2\beta = 90^\circ$$

$$\Rightarrow \alpha + \beta = 45^\circ$$

$$\Rightarrow \angle MCN = 45^\circ$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (B,C)

$$\begin{aligned}\int \sqrt{\sec^2 x + 3} dx &= \int \frac{\sec^2 x + 3}{\sqrt{\sec^2 x + 3}} dx \\&= \int \frac{\sec^2 x}{\sqrt{\tan^2 x + 4}} dx + \int \frac{3 \cos x dx}{\sqrt{4 - 3 \sin^2 x}} \\&= \ln \left(\tan x + \sqrt{\tan^2 x + 4} \right) + \sqrt{3} \sin^{-1} \left(\frac{\sqrt{3}}{2} \sin x \right) + c\end{aligned}$$

$$f(x) = \ln \left(\tan x + \sqrt{\tan^2 x + 4} \right)$$

$$g(x) = \sin^{-1} \left(\frac{\sqrt{3}}{2} \sin x \right) + 1$$

6. Ans (A,B)

$$I = \int_{-\pi/4}^{\pi/4} \frac{e^{\sec x} |\sin x| \sec^2 x}{(1 + e^{\sin x})} dx$$

apply king & add

$$2I = \int_{-\pi/4}^{\pi/4} e^{\sec x} |\sin x| \sec^2 x dx$$

$$I = \int_0^{\pi/4} e^{\sec x} \cdot \sec x \cdot \tan x dx$$

$$e^{\sec x} = t$$

$$I = \int_e^{e^{\sqrt{2}}} dt = e^{\sqrt{2}} - e$$

7. Ans (B)

$$\lim_{x \rightarrow 5} \frac{\int_1^{f^2(x)} (x+t) e^t dt}{x-5}$$

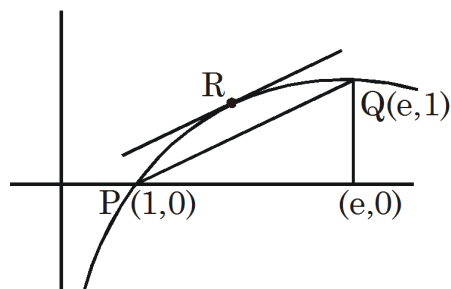
$$\lim_{x \rightarrow 5} \frac{x \int_1^{f^2(x)} e^t dt + \int_1^{f^2(x)} t e^t dt}{x-5}$$

use L hospital

$$\lim_{x \rightarrow 5} \left[x e^{f^2(x)} 2f(x) f'(x) + f^2(x) e^{f^2(x)} 2f(x) f'(x) + \int_1^{f^2(x)} e^t dt \right]$$

$$5e'2 \left(\frac{1}{12e} \right) + e'2 \left(\frac{1}{\sqrt{2}e} \right) + 0 = 1$$

8. Ans (A,C,D)



Let R point is (x_1, y_1) then

$$\frac{dy}{dx} \bigg|_{(x_1, y_1)} = \text{Slope of PQ}$$

$$\frac{1}{x_1} = \frac{1}{e-1} \text{ or } \boxed{x_1 = e-1}$$

Now for option (C)

Area under tangent > Area under curve

for option (D)

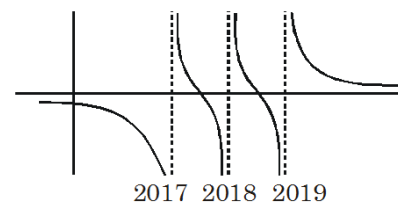
Equation of normal at 'P' (1, 0) is $x + y = 1$

which is passing at (0, 1)

9. Ans (A,B,C,D)

$$f(x) = x$$

$$g(x) = \frac{1}{x-2017} + \frac{1}{x-2018} + \frac{1}{x-2019}$$



$$g'(x) = - \left(\frac{1}{(x-2017)^2} + \frac{1}{(x-2018)^2} + \frac{1}{(x-2019)^2} \right)$$

$$g'(2016) = - \left(1 + \frac{1}{4} + \frac{1}{9} \right) = \frac{-49}{36}$$

10. Ans (A,B,C)

$$\text{Let } h(x) = f(x) - 5g(x)$$

$h(x)$ is continuous, differentiable & $f'(x)$ is also

continuous, differentiable

$$h(-2) = 6 = h(0) = h(1) = h(3) \text{ \& } (f-5g)''$$

never vanishes

$\Rightarrow$ exactly one root in $(-2, 0)$, $(0, 1)$ & $(1, 3)$

11. Ans (A,B,C)

$$\lim_{\theta \rightarrow \frac{\pi}{2}^-} \frac{1}{\tan^9 \theta} \left((1 + \tan \theta)^{10} + (2 + \tan \theta)^{10} + \dots + (20 + \tan \theta)^{10} \right) - 20 \tan^9 \theta$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}^-} \frac{(1 + \tan \theta)^{10} + \dots + (20 + \tan \theta)^{10} - 20 \tan^{10} \theta}{\tan^9 \theta}$$

$$\frac{((\tan^{10} \theta + {}^{10}C_1 \tan^9 \theta) + (\tan^{10} \theta + {}^{10}C_1 2 \tan^9 \theta) + \dots + (\tan^{10} \theta + {}^{10}C_1 \cdot 20 \cdot \tan^9 \theta)) - 20 \tan^{10} \theta}{\tan^9 \theta}$$

$$\Rightarrow {}^{10}C_1 (1 + 2 + 3 + \dots + 20) = \frac{10 \cdot 20 \cdot 21}{2}$$

$$\Rightarrow 2100$$

12. Ans (A,B,C)

$$\text{Clearly } f_1(x) = \frac{\pi}{2} - \sin^2 x$$

$$f_2(x) = \frac{\pi}{2} - \cos^2 x$$

$$f_3(x) = \frac{\pi}{2} - \cos^2 x$$

$$f_4(x) = \frac{\pi}{2} - \sin^2 x$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (16.00)

$$y = 3 + \frac{mx + n - 3}{1 + x^2}$$

now $mx + n - 3 < 0$ Think !

$$m = 0 \quad \forall x \in \mathbb{R}$$

$$n - 3 < 0$$

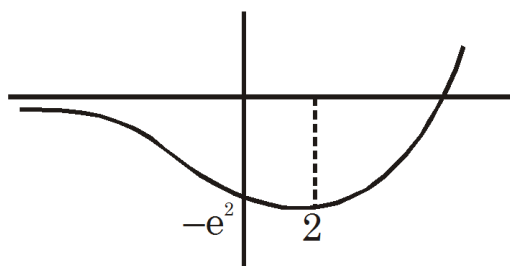
$$y = 3 + \frac{n - 3}{1 + x^2}$$

$$y_{\min} \text{ at } x = 0 \quad \text{think !}$$

$$y = 3 + n - 3 = -4$$

$$n = -4$$

2. Ans (7.00)



$$e^x(x-3) = k$$

$$\text{Let } f(x) = e^x(x-3)$$

$$f'(x) = e^x(x-2)$$

$$k \in (-e^2, 0)$$

3. Ans (2.00)

$$\int \frac{(3x^2 + 2x)dx}{x^6 + 2x^5 + x^4 + 2x^3 + 2x^2 + 5}$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{x^3 + x^2 + 1}{2} \right) + C$$

$$\therefore a = 2 \text{ and } g(1) = \frac{3}{2}, \quad g(0) = \frac{1}{2}$$

4. Ans (0.11)

$$x = 1, \quad f(1) + 2f(1) = 1$$

$$\Rightarrow f(1) = \frac{1}{3}$$

$$f(x^2) + 2x^2 f'(x^2) + (x^2 + 1)f'(x) + 2xf(x) = 1$$

$$x = 1$$

$$f(1) + 2f'(1) + 2f'(1) + 2f(1) = 1$$

$$4f'(1) + 1 = 1 \Rightarrow f'(1) = 0$$

$$2xf'(x^2) + 4x^3 f''(x^2) + 4xf'(x^2) + 2xf'(x) +$$

$$(x^2 + 1)f''(x) + 2f(x) + 2f'(x) = 0$$

$$x = 1$$

$$2f'(1) + 4f''(1) + 4f'(1) + 2f'(1) + 2f''(1) +$$

$$2f(1) + 2f'(1) = 0$$

$$6f''(1) + \frac{2}{3} = 0$$

$$f''(1) = -\frac{1}{9}$$

5. Ans (5.00)

$$f(x+1) = e^{x^2} f(x)$$

$$\log f(x+1) = x^2 + \log f(x)$$

$$\Rightarrow \log f(x+1) - \log f(x) = x^2$$

$$\Rightarrow \frac{f'(x+1)}{f(x+1)} - \frac{f'(x)}{f(x)} = 2x$$

$$\Rightarrow \frac{f'(x + \frac{1}{2})}{f(x + \frac{1}{2})} - \frac{f'(x - \frac{1}{2})}{f(x - \frac{1}{2})} = 2x - 1$$

$$\text{put } x = 1, 2, \dots, N \text{ and adding}$$

$$\Rightarrow \frac{f'(N + \frac{1}{2})}{f(N + \frac{1}{2})} - \frac{f'(\frac{1}{2})}{f(\frac{1}{2})} = 1 + 3 + 5 + \dots + (2N - 1)$$

$$\Rightarrow \frac{f'(N + \frac{1}{2})}{f(N + \frac{1}{2})} - 5 = N^2$$

6. Ans (4.00)

$$\sin^{-1} x + \sin^{-1} y \geq (x+y)^2 - 2[xy + x - 1 + y] + \pi$$

$$\sin^{-1} x + \sin^{-1} y \geq x^2 + y^2 - 2x - 2y + 1 + 1 + \pi$$

$$\left(\frac{\pi}{2} - \sin^{-1} x\right) + \left(\frac{\pi}{2} - \sin^{-1} y\right) + (x-1)^2 + (y-1)^2 \leq 0 \dots (i)$$

All terms on LHS are ≥ 0 , so (i) is

valid when $\sin^{-1} x = \sin^{-1} y = \frac{\pi}{2}$; $x = y = 1$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

27-06-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS & TNAS)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	B,C	A,D	C,D	A,B,C	A,C,D	B,C	C	B,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A	D	B	A				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	2.00	20.00	1.00	10.00	12.00	90.00		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C	A,B,C	B,C,D	A,B,C,D	A,B,C,D	B,D	A,B	A
SECTION-I (ii)	Q.	9	10	11	12				
	A.	B	B	D	C				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	13.02	8.00	4.00	7.00	5.00	12.00		

PART-3 : MATHEMATICS

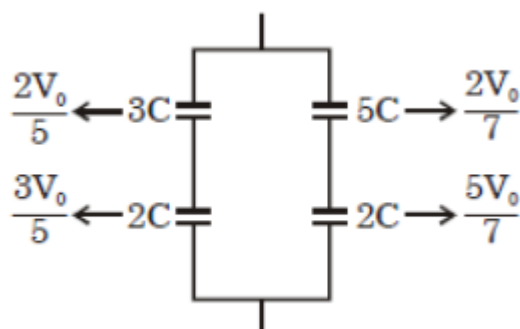
SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	B,C,D	A,B	A,D	A,B,D	A,B	A,C,D	A,C	A,B,C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C	D	B	C				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	2.00	2.00	4.50	3.00	3.50	1.50		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B,C)



Now, we can compare electric fields easily.

2. Ans (A,D)

Total charge on the three plates connected to A is $= 2 \times 5 + 3 \times 20 + 5 \times 10 = 120 \mu\text{C}$.

After closing the switch, the sum of charges on the 4 plates shown inside box is still zero. This is because the total charge supplied by the cell to the system of 4 plates is zero.

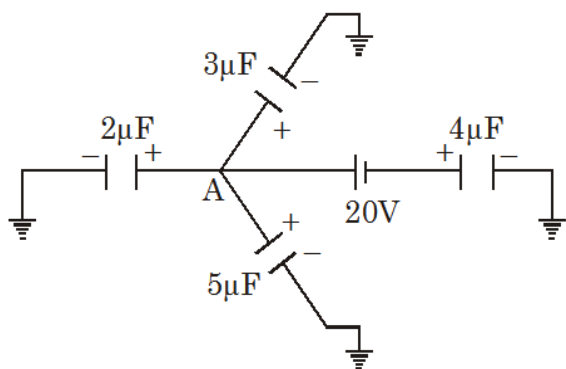
Let potential of point A be V_0 .

Charge on $2 \mu\text{F}$ capacitor $= 2 V_0$

Charge on $3 \mu\text{F}$ capacitor $= 3 V_0$

Charge on $5 \mu\text{F}$ capacitor $= 5 V_0$

Charge on $4 \mu\text{F}$ capacitor $= 4 (V_0 - 20)$



$$2V_0 + 3V_0 + 5V_0 + 4(V_0 - 20) = 120$$

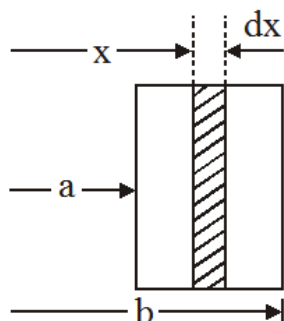
$$\text{or, } V_0 = \frac{200}{14} = \frac{100}{7} \text{ V}$$

$$\text{Charge on } 2 \mu\text{F capacitor is } = 2V_0 = \frac{200}{7} \mu\text{C}$$

Sum of charge on the plates of the capacitors facing A has not changed. Therefore, sum of charges on the 4 plates connected to the ground is still $-120 \mu\text{C}$. The total charge flow to the ground is zero.

3. Ans (C,D)

Consider a strip of width dx shown on the rectangular face in the figure



Think of a half cylindrical conductor of radius x and infinitesimally small thickness dx .

Length of this conductor $= \pi x$

Cross sectional area of this conductor $= hdx$

$\therefore$ Resistance of this thin cylindrical conductor

$$dR = \frac{\rho \pi x}{h dx}$$

The given conductor is made of countless number of such thin conductors, all connected in parallel.

$$\therefore \frac{1}{R} = \int \frac{1}{dR} = \frac{h}{\rho \pi} \int_a^b \frac{dx}{x}$$

$$= \frac{h}{\rho \pi} \ln \left(\frac{b}{a} \right)$$

$$\therefore R = \frac{\rho \pi}{h \ln \left(\frac{b}{a} \right)}$$

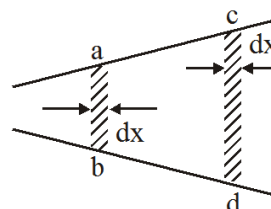
let area at $ab = A$

And area at $cd = 2A$

Current in the wire $= I$. Current in same at all cross sections.

$$\text{Current density at } ab \quad J_1 = \frac{I}{A}$$

$$\text{Current density at } cd \quad J_2 = \frac{I}{2A} = \frac{J_1}{2}$$



Consider two discs of equal thickness dx at section ab and cd .

$$\text{Resistance of disc at } ab \quad dR_1 = \sigma \frac{dx}{A}$$

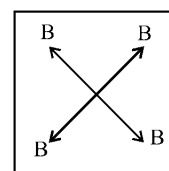
$$\text{Rate of Heat dissipation in disc at } ab = I^2 (dR_1)$$

$$\text{Rate of heat dissipation per unit volume} = \frac{I^2 (dR_1)}{A \cdot dx}$$

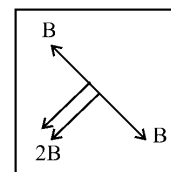
4. Ans (A,B,C)

1st and 4th case

$$B_{\text{net}} = 0$$

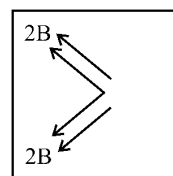


2nd case



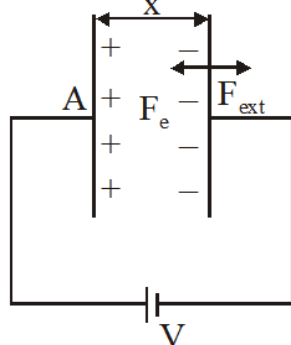
$$B_{\text{net}} = 2B = 2 \left[\frac{\mu_0 i}{2\pi \left(\frac{a}{\sqrt{2}} \right)} \right]$$

3rd case



$$B_{\text{net}} = 4B \cos 45^\circ = 2\sqrt{2} \left[\frac{\mu_0 i}{2\pi \left(\frac{a}{\sqrt{2}} \right)} \right]$$

5. Ans (A, C, D)



(a) Charge on capacitor is $Q = CV$ $\left[C = \frac{\epsilon_0 A}{d} \right]$

$$W_{\text{battery}} = Q \cdot V = CV^2$$

$$\text{Energy stored in the capacitor } U = \frac{1}{2} CV^2$$

$$\therefore \text{Heat generated} = W_{\text{batt}} - U = \frac{1}{2} CV^2$$

(b) Let the distance between the plates be x at some intermediate stage.

$$C = \frac{\epsilon_0 A}{x}$$

$$q = CV = \frac{\epsilon_0 AV}{x}$$

$$\text{Field due to charge on one plate is } E = \frac{q}{2A\epsilon_0}$$

$$\therefore \text{Force between plates } F = Eq = \frac{q^2}{2A\epsilon_0} = \frac{\epsilon_0 AV^2}{2x^2}$$

(c) Assuming that the positive plate is fixed and the negative plate moves slowly towards it, work done by external agent in displacement dx is

$$dW_{\text{ext}} = -|F_{\text{ext}}| |dx| = -|F_e| |dx| = \frac{\epsilon_0 AV^2}{2} \frac{dx}{x^2}$$

$\therefore$ Work done in reducing the distance from d to $\frac{d}{2}$ is

$$W_{\text{ext}} = + \frac{\epsilon_0 AV^2}{2} \int_d^{d/2} \frac{dx}{x^2} = \frac{\epsilon_0 AV^2}{2d} = -\frac{1}{2} CV^2$$

(c) Energy dissipated must be zero as the plates are moved slowly. Charge flow from battery to the capacitor plates is very slow.

$$W_{\text{ext}} = \frac{1}{2} CV^2$$

Change in PE stored in the capacitor

$$\Delta U = \frac{1}{2} 2CV^2 - \frac{1}{2} CV^2 = \frac{1}{2} CV^2$$

[Capacitance doubled when d was halved]

$$W_{\text{batt}} = [(2CV) - CV] V = CV^2$$

$$\therefore W_{\text{batt}} + W_{\text{ext}} = \Delta U + \text{Heat generated}$$

$$\therefore \text{Heat generated} = 0$$

8. Ans (B, C)

Let speed of both be V just before collision

Energy conservation

$$\frac{1}{2} mV^2 \times 2 + \frac{K(3q)(-q)}{2r} = \frac{K(3q)(-q)}{R}$$

$$\therefore mV^2 = 3Kq^2 \left[\frac{1}{2r} - \frac{1}{R} \right] \dots (i)$$

In elastic collision, exchange of velocity will take place and charge on both balls will now be $\frac{3q - q}{2} = q$. Now, the two balls repel each other. Speed will be maximum when separation between the centers of the balls become $2R$.

$$\frac{1}{2} mV_0^2 \times 2 + \frac{Kq^2}{2R} = \frac{1}{2} mV^2 \times 2 + \frac{Kq^2}{2r}$$

$$mV_0^2 = 3Kq^2 \left[\frac{1}{2r} - \frac{1}{R} \right] + \frac{Kq^2}{2} \left[\frac{1}{r} - \frac{1}{R} \right]$$

$$= \frac{Kq^2}{2rR} [3R - 6r + R - r] = \frac{Kq^2}{2rR} (4R - 7r)$$

$$\therefore V_0 = \sqrt{\frac{Kq^2}{2mrR} (4R - 7r)}$$

Note : Energy loss takes places when charge redistribution takes place during collision.

PART-1 : PHYSICS

SECTION-I (ii)

9. Ans (A)

The charge in the cavity can change the magnitude of charge appearing on inner and outer surface of sphere and charge distribution on the inner surface but it can't change the charge distribution on the outer surface of sphere. Charge placed at a point outside the sphere can only change the charge distribution of the outer surface.

10. Ans (D)

The charge in the cavity can change the magnitude of charge appearing on inner and outer surface of sphere and charge distribution on the inner surface but it can't change the charge distribution on the outer surface of sphere. Charge placed at a point outside the sphere can only change the charge distribution of the outer surface.

11. **Ans (B)**

For geo-stationary satellite, time period of satellite equals to time period of rotation of planet about its axis.

Also for it,

$$\frac{GMm}{(R+h)^2} = \frac{mv^2}{(R+h)} \Rightarrow v \propto \frac{1}{\sqrt{R+h}}$$

Any process changing ω_{planet} will change the height of satellite for it to remain geo-stationary.

12. **Ans (A)**

For geo-stationary satellite, time period of satellite equals to time period of rotation of planet about its axis.

Also for it,

$$\frac{GMm}{(R+h)^2} = \frac{mv^2}{(R+h)} \Rightarrow v \propto \frac{1}{\sqrt{R+h}}$$

Any process changing ω_{planet} will change the height of satellite for it to remain geo-stationary.

PART-1 : PHYSICS

SECTION-II

1. **Ans (2.00)**

$$\frac{1}{2}(1.5v_0)^2 - \frac{GMm}{r_0} = \frac{1}{2}mv^2 \quad \dots\dots(1)$$

$$v_0 = \sqrt{\frac{GM}{r_0}} \quad \dots\dots(2)$$

$$\therefore v = \frac{v_0}{2}$$

2. **Ans (20.00)**

$$\text{Initially } \frac{R_1}{R_2} = \frac{2}{3} = \frac{\ell}{100 - \ell} \Rightarrow \ell = 40 \text{ cm}$$

$$\text{Now } \frac{R_2}{R_1} = \frac{3}{2} = \frac{\ell}{100 - \ell} \Rightarrow \ell = 60 \text{ cm}$$

shift = 20 cm

3. **Ans (1.00)**

$$q = q_0 \left(1 - e^{-\frac{1}{\tau}} \right)$$

$$q_0 = \frac{CV}{2}$$

$$\tau = R_{\text{eq}} C_{\text{eq}} = \frac{CR}{2}$$

$$\therefore q = \frac{CV}{2} \left(1 - e^{-\frac{2t}{RC}} \right)$$

$$\therefore \alpha = 1$$

4. **Ans (10.00)**

$$\text{Given } \lambda_0 T_0 = b$$

Let find temp. of sphere becomes $(T_0 + \Delta T)$

$$\therefore 0.4 \sigma S e T_0^4 = \sigma e S (T^4 - T_0^4)$$

$$\Rightarrow 0.4 T_0^4 = (T_0 + \Delta T)^4 - T_0^4$$

$$\Rightarrow 0.4 T_0^4 = T_0^4 \left[\left(1 + \frac{\Delta T}{T_0} \right)^4 - 1 \right]$$

$$\Rightarrow 0.4 = 1 + \frac{4\Delta T}{T_0} - 1 \Rightarrow \Delta T = 0.1 T_0$$

$$\therefore T_f = 1.1 T_0$$

$$\therefore \lambda_0 T_0 = \lambda' 1.1 T_0$$

$$\Rightarrow \lambda = \frac{10}{11} \lambda_0$$

5. **Ans (12.00)**

$$\text{Heat gained by ice} = m S_{\text{ice}} \times 10 + m S_W \times 20 + m L_f$$

$$= 105 \text{ m cal}$$

$$\text{Heat lost by calorimeter} = 10 \times 1 \times 20 = 200 \text{ cal}$$

$$\text{Heat lost by liq.} = -50 \int_{40}^{20} \left(\frac{1+\theta}{500} \right) d\theta = 50 \left[\theta + \frac{\theta^2}{1000} \right]_{20}^{40}$$

$$= 50 \left[\left(40 + \frac{1600}{1000} \right) - \left(20 + \frac{400}{1000} \right) \right]$$

$$= 50 \times 21.2 = 1060 \text{ cal}$$

$$\therefore 105 m = 1060 + 200 \Rightarrow m = 12 \text{ g}$$

6. **Ans (90.00)**

Given, $f_L = 30 \text{ cm}$

$$\therefore \frac{1}{30} = \left(\frac{3}{2} - 1 \right) \left(\frac{2}{R} \right) \Rightarrow R = 30 \text{ cm}$$

$$\therefore \frac{1.5}{v_1} + \frac{1}{90} = \frac{1.5 - 1}{30} \Rightarrow v_1 = 270 \text{ [For air-lens refraction]}$$

Again, (for lens water refraction)

$$\frac{4}{3v_2} - \frac{3}{2 \times 270} = \frac{\frac{4}{3} - \frac{3}{2}}{-30}$$

$$\Rightarrow \frac{4}{3v_2} = \frac{1}{180} + \frac{1}{180} = \frac{1}{90}$$

$$\therefore v_2 = 120 \text{ cm}$$

$\therefore$ Image formed by mirror will be in front of it at a distance 40 cm.

Now,

$$\frac{3}{2v_3} + \frac{4}{3 \times 40} = \frac{\frac{3}{2} - \frac{4}{3}}{30} \text{ (For water lens refraction)}$$

$$\Rightarrow \frac{3}{2v_3} = \frac{1}{180} - \frac{1}{30} = \frac{1 - 6}{180} = \frac{-5}{180}$$

$$\therefore v_3 = \frac{-270}{5}$$

$\therefore$ For final image, (for lens-air refraction)

$$\frac{1}{v_f} + \frac{3 \times 5}{2 \times 2.70} = \frac{1 - 1.5}{-30}$$

$$\Rightarrow \frac{1}{v_f} = \frac{1}{60} - \frac{5}{180} = \frac{3 - 5}{180} = \frac{-2}{180}$$

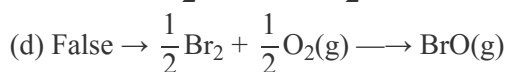
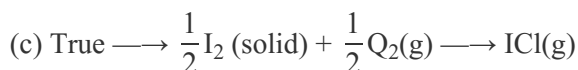
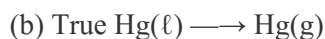
$$\therefore v_f = -90 \text{ cm}$$

PART-2 : CHEMISTRY

SECTION-I (i)

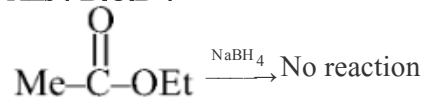
2. **Ans (A,B,C)**

Enthalpy of formation is defined as enthalpy change when 1 mole of compounds is made from its constituent elements.



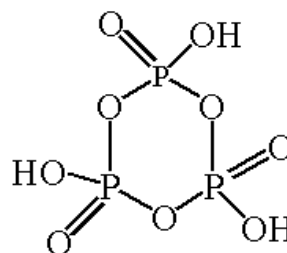
Not in standard state i.e. $\text{Br}_2(\ell)$

3. **Ans (B,C,D)**



6. **Ans (B,D)**

Structure of tricyclic metaphosphoric acid (HPO_3) is as follows :



PART-2 : CHEMISTRY

SECTION-I (ii)

11. **Ans (D)**

I : $[\text{MnO}_4^-]$ d^3s , diamagnetic, colored complex

II : $[\text{Ni}(\text{dmg})_2]$ dsp^2 , diamagnetic, colored complex

12. **Ans (C)**

III : $[\text{FeO}_4]^{2-}$ d^3s , paramagnetic, colored complex

IV : $[\text{Zn}(\text{gly})_2]$ sp^3 , diamagnetic, colourless

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (13.02)**

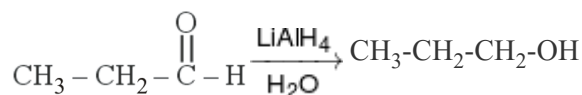
$$\ln \frac{k_2}{k_1} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\ln 2 = \frac{E_a}{2} \left(\frac{1}{300} - \frac{1}{310} \right)$$

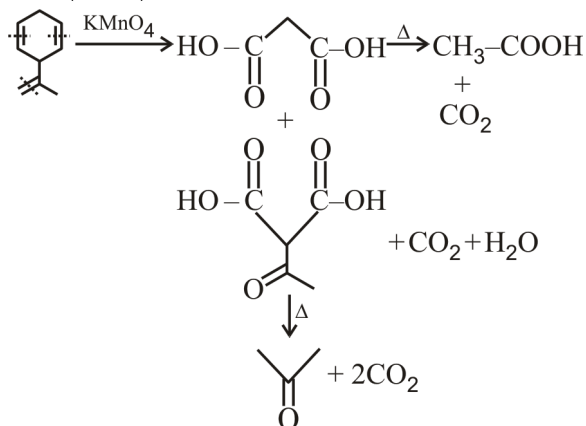
$$E_a = 13020 \text{ cal}$$

2. **Ans (8.00)**

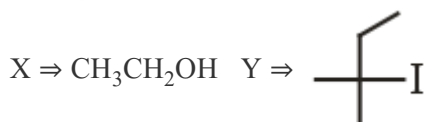
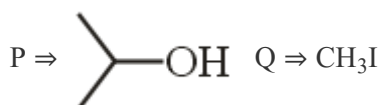
i, ii, iii, iv, v, vi, vii, viii



3. Ans (4.00)



4. Ans (7.00)



5. Ans (5.00)

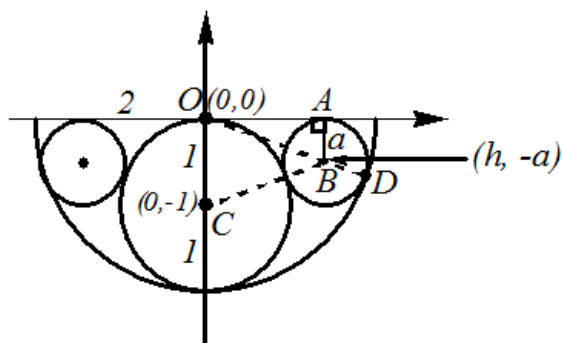
2,3,4,5,6

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B,C,D)

From figure, there are two possible circles of radius a.



$$BC = \sqrt{h^2 + (a-1)^2} = a+1 \quad \text{and}$$

$$OB = \sqrt{h^2 + a^2} = 2-a.$$

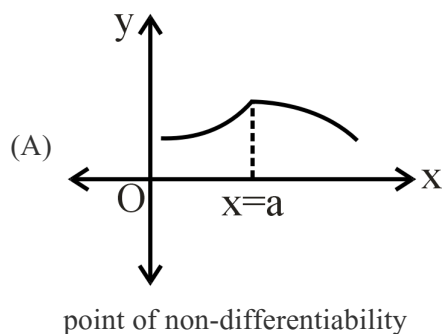
Solving we get $h = \pm\sqrt{2}$, $a = \frac{1}{2}$. To get point D

solve equation

of line OBD with lower semicircle $y = -\sqrt{4-x^2}$.

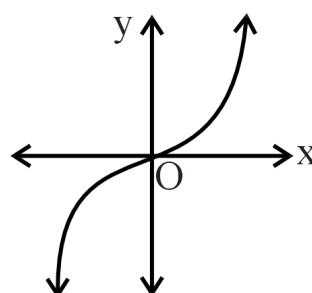
D can be $\left(-\frac{4\sqrt{2}}{3}, \frac{-2}{3}\right)$ or $\left(\frac{4\sqrt{2}}{3}, \frac{-2}{3}\right)$

2. Ans (A,B)



(B) $y = x^3$

$$f'(0) = 0$$

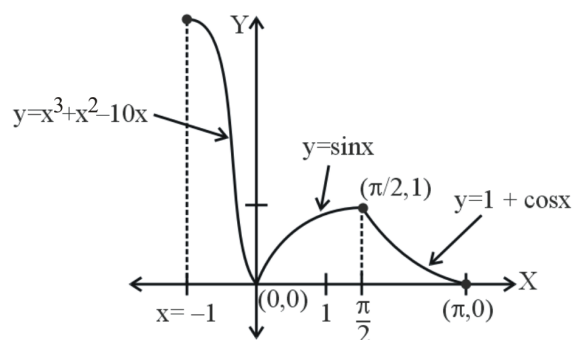


But at $x = 0$, $f(x)$ is increasing

(C) $f''(c)$ can be zero also e.g. $f(x) = -x^4$.

(D) false e.g. $f(x) = x^3$ at $x = 0$ is increasing

3. Ans (A,D)



Graph of $f(x)$

Clearly, from above graph, $f(x)$ has local maximum at $x = \frac{\pi}{2}$ and absolute maximum at

$x = -1$

4. **Ans (A,B,D)**

Given, $f(t) = e^t(\cos^2 t - \sin 2t)$

∴ Integrate both sides with respect to t, we get

$$f(t) = e^t \cos^2 t + C$$

But, $f(0) = 1 \Rightarrow C = 0$

So, $f(t) = e^t \cos^2 t$

Clearly,

[As, $-\infty < t < 0$ so $0 < e^t < 1$ and $0 \leq \cos^2 t \leq 1$

$\Rightarrow e^t \cos^2 t \leq 1 \Rightarrow f$ is bounded in $n(-\infty, 0)$.]

Note that f is neither odd nor even function.

Also, $f(t) = e^t \Rightarrow e^t \cos^2 t = e^t \Rightarrow \cos^2 t = 1$

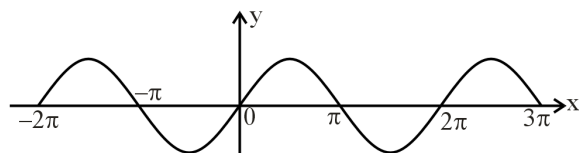
∴ $t = 0, 2\pi \in [0, 2\pi]$. So, 3 solutions exist.

Also, $\lim_{t \rightarrow 0} (f(t))^{\frac{1}{t}} (1^\infty \text{ form}) = e^L$, where

$$L = \lim_{t \rightarrow 0} \left(\frac{e^t \cos^2 t - 1}{t} \right) \quad \left(\frac{0}{0} \text{ form} \right)$$

$$= \lim_{t \rightarrow 0} \frac{e^t (\cos^2 t - \sin 2t)}{1} = e^t \quad (\text{Using L'hospital rule})$$

5. **Ans (A,B)**



$$f'(x) = \frac{\sin x}{(1 + \cos^2 x)} 2x = 0, x = 0, n\pi$$

$x = -\pi, 0, \pi, 2\pi$

$\lim_{x \rightarrow 2\pi^+} f'(x) = \text{positive quantity}$

$\lim_{x \rightarrow 2\pi^-} f'(x) = \text{Negative, so maxima at } x = 2\pi$

$\lim_{x \rightarrow \pi^+} f'(x) = -ve, \lim_{x \rightarrow \pi^-} f'(x) = +ve$, so

maxima

$\lim_{x \rightarrow 0^+} f'(x) = +ve, \lim_{x \rightarrow 0^-} f'(x) = +ve$, Neither

maxima, nor minima

$\lim_{x \rightarrow -\pi^+} f'(x) = +ve, \lim_{x \rightarrow -\pi^-} f'(x) = -ve$, minima

6. **Ans (A,C,D)**

We have

$$g(x) = \begin{cases} -\pi \cos\left(\frac{\pi}{x}\right) + 2x \sin\left(\frac{\pi}{x}\right) - \pi \cos\left(\frac{\pi}{x-1}\right) + 2(x-1) \sin\left(\frac{\pi}{x-1}\right) & ; x \neq 0, 1 \\ 0 & ; x = 0, 1 \end{cases}$$

Clearly, $g'(x)$ is discontinuous at both $x = 0$ and $x = 1$.

Also, $g(0) = 0 = g(1)$ (Given)

So, Rolle's theorem is applicable for $g(x)$ in $[0, 1]$

7. **Ans (A,C)**

For continuous : $3 = k + 4 \Rightarrow k = -1$

Derivability depends on continuity and value of k.

For local minima at $x = 2$: $k + 4 > 3$
 $\Rightarrow k > -1$

8. **Ans (A,B,C,D)**

(A) $\lim_{x \rightarrow 1} \left[\frac{f(x^3) - f(1)}{(x^3 - 1)} \cdot (x^2 + x + 1) \right]$
 $= \lim_{x \rightarrow 1} \left(\frac{f(1 + (x^3 - 1)) - f(1)}{x^3 - 1} \right) \times 3 = 3f'(1)$

(B) $\lim_{x \rightarrow 0} \left[\frac{f(ax) - f(0)}{ax} \cdot a \frac{-(f(bx) - f(0))}{bx} \cdot b \right] \cdot \frac{1}{c}$
 $= \frac{a-b}{c} f'(0)$

(C) $\lim_{x \rightarrow x_0} \frac{xf(x_0) - x_0f(x)}{x - x_0}$

Put $x = x_0 + h$

$$\lim_{h \rightarrow 0} \frac{(x_0 + h)f(x_0) - x_0f(x_0 + h)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x_0(f(x_0) - f(x_0 + h)) + hf(x_0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-x_0(f(x_0 + h) - f(x_0))}{h} + f(x_0)$$

$$= -x_0f'(x_0) + f(x_0) = f(x_0) - x_0f'(x_0)$$

(D) $\frac{f(h) - f(0)}{h} < 0$ and $\frac{f(-h) - f(0)}{-h} > 0$
 $h > 0$

$$\Rightarrow f'(0^+) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} \leq 0,$$

$$f'(0^-) = \lim_{h \rightarrow 0} \frac{f(-h) - f(0)}{-h} \geq 0$$

But $f'(0^+) = f'(0^-) \Rightarrow f'(0) = 0$

PART-3 : MATHEMATICS

SECTION-I (ii)

9. Ans (C)

(I) Point is (1,0)

$$\left. \frac{dy}{dx} \right|_{(1,0)} = -1$$

$\Rightarrow$ Tangent is $x + y = 1$

(II) Apply L.M.V.T

$$f'(c) = \frac{f(9) - f(2)}{7}$$

$$\Rightarrow -4 \leq \frac{f(9) - f(2)}{7} \leq 3$$

$$\Rightarrow -28 \leq f(9) - f(2) \leq 21$$

$$(III) y = \frac{x^4}{12} + 16\ln x - 16\ln 2 - \frac{10}{3}$$

$$\frac{dy}{dx} = \frac{4x^3}{12} + \frac{16}{x}$$

$$\frac{d^2y}{dx^2} = \frac{12x^2}{12} - \frac{16}{x^2} = \frac{x^4 - 16}{x^2}$$

$$\frac{d^2y}{dx^2} = 0 \Rightarrow x = 2 \Rightarrow \alpha = 2 \text{ \& } \beta = -2$$

$$(IV) f'(x) = 2x^2 + 3x - \lambda \log x$$

$$f''(x) = 4x + 3 - \frac{\lambda}{x} = \frac{4x^2 + 3x - \lambda}{x} < 0$$

$$\Rightarrow 4x^2 + 3x - \lambda < 0 \text{ in } (0,1)$$

$$\Rightarrow \lambda > 4x^2 + 3x$$

$$\Rightarrow \lambda > 7$$

10. Ans (D)

(I) Point is (1,0)

$$\left. \frac{dy}{dx} \right|_{(1,0)} = -1$$

$\Rightarrow$ Tangent is $x + y = 1$

(II) Apply L.M.V.T

$$f'(c) = \frac{f(9) - f(2)}{7}$$

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$$\Rightarrow \lambda > 4x^2 + 3x$$

$$\Rightarrow \lambda > 7$$

11. Ans (B)

Case-I : If $a > 0$, then

$$gof(x) = \begin{cases} x + a + 1; & x < -a \\ (x + a)^2 + 2(b - x - a) + 1; & -a \leq x < 0 \\ (1 - x)^2 + 2(b - 1 + x) + 1; & 0 \leq x < 1 \\ (x - 1)^2 + 2(b - x + 1) + 1; & x \geq 1 \end{cases}$$

$$a = 1 \text{ \& } b = 0$$

Case-II : If $a = 0$, then

$$g(f(x)) = \begin{cases} x + 1 & x < 0 \\ x^2 + 2b & 0 \leq x < 1 \\ (x - 2)^2 + 2b & x \geq 1 \end{cases}$$

$$a = 0; b = \frac{1}{2}$$

$$\left. \frac{d(g(f(x)))}{dx} \right|_{x=4} = (2(x - 1) - 2)|_{x=4} = 4$$

12. Ans (C)

Case-I : If $a > 0$, then

$$gof(x) = \begin{cases} x + a + 1; & x < -a \\ (x + a)^2 + 2(b - x - a) + 1; & -a \leq x < 0 \\ (1 - x)^2 + 2(b - 1 + x) + 1; & 0 \leq x < 1 \\ (x - 1)^2 + 2(b - x + 1) + 1; & x \geq 1 \end{cases}$$

$$a = 1 \text{ \& } b = 0$$

Case-II : If $a = 0$, then

$$g(f(x)) = \begin{cases} x + 1 & x < 0 \\ x^2 + 2b & 0 \leq x < 1 \\ (x - 2)^2 + 2b & x \geq 1 \end{cases}$$

$$a = 0; b = \frac{1}{2}$$

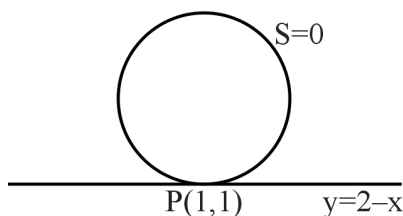
$$\left. \frac{d(g(f(x)))}{dx} \right|_{x=4} = (2(x - 1) - 2)|_{x=4} = 4$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (2.00)

Equation of S is $(x - 1)^2 + (y - 1)^2 + \lambda(x + y - 2) = 0$



$$x^2 + y^2 + x(\lambda - 2) + y(\lambda - 2) + 2(1 - \lambda) = 0 \dots(1)$$

$$x^2 + y^2 + 2x + 2y - 2 = 0 \dots(2)$$

Given (1) and (2) are orthogonal, so

$$\frac{2(\lambda - 2)}{2}(1) + 2\left(\frac{\lambda - 2}{2}\right)(1) = 2(1 - \lambda) - 2$$

$$\Rightarrow \lambda - 2 + \lambda - 2 = -2\lambda$$

$$4\lambda = 4 \Rightarrow \lambda = 1$$

Hence equation of S is $x^2 + y^2 - x - y = 0$

Now, length of tangent from (2, 2) is

$$\sqrt{4 + 4 - 2 - 2} = 2$$

2. Ans (2.00)

$$I = \int_{1/2}^1 \frac{\tan^{-1} x}{x} dx + \int_1^2 \frac{\tan^{-1} x}{x} dx$$

$$= \int_1^2 \frac{\tan^{-1} \frac{1}{x}}{x} dx + \int_1^2 \frac{\tan^{-1} x}{x} dx = \frac{\pi}{2} \ln 2$$

3. Ans (4.50)

Given,

$$f(x) = e^x \cdot \int_0^x e^{-y} \cdot f'(y) dy - (x^2 - x + 1)e^x$$

...(1)

Differentiating both the sides

$$f'(x) = e^x \cdot e^{-x} \cdot f'(x) +$$

$$e^x \int_0^x e^{-y} \cdot f'(y) dy - [(x^2 - x + 1)e^x + e^x(2x - 1)]$$

$$f'(x) = f'(x) + \int_0^x e^{x-y} \cdot f'(y) dy - e^x (x^2 + x)$$

$$\Rightarrow 0 = f(x) + (x^2 - x + 1)e^x - e^x(x^2 + x) \text{ [Substituting]}$$

$$\int_0^x e^{x-y} f'(y) dy = f(x) + (x^2 - x + 1)e^x \text{ from (1)}$$

$$f(x) = e^x(x^2 + x - x^2 + x - 1)$$

$$f(x) = e^x(2x - 1) \Rightarrow f(1) = e$$

$$\text{Also, } f'(x) = e^x \cdot 2 + (2x - 1)e^x \Rightarrow f'(1) = 2e + e = 3e$$

$$\text{and } f''(x) = 2e^x + (2x - 1)e^x + 2e^x$$

$$\Rightarrow f''(1) = 2e + e + 2e = 5e$$

$$\text{Hence } f(1) + f'(1) + f''(1) = 9e \Rightarrow k = 9$$

4. Ans (3.00)

We have $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$f'(x) = \lim_{h \rightarrow 0} \frac{f\left(\frac{2x+2h}{2}\right) - f\left(\frac{2x+2 \times 0}{2}\right)}{h}$

$= \lim_{h \rightarrow 0} \frac{\frac{f(2x)+f(2h)}{2} - \frac{f(2x)+f(0)}{2}}{h}$

$= \lim_{h \rightarrow 0} \frac{f(2h) - f(0)}{2h} = f'(0) = k \text{ (say)}$

$f(x) = k$

$f(x) = kx + c ; f(0) = 0 \Rightarrow c = 0$

$\therefore f(x) = kx$

Now,

$I = \int_0^{2\pi} (kx - \sin x)^2 dx = \int_0^{2\pi} (k^2x^2 - 2kx \sin x + \sin^2 x) dx$

$= k^2 \cdot \left[\frac{x^3}{3} \right]_0^{2\pi} - 2k \int_0^{2\pi} x \sin x dx + \int_0^{2\pi} \sin^2 x dx$

$= k^2 \cdot \frac{8\pi^3}{3} - 2k(-2\pi) + \pi = \frac{8k^2\pi^3}{3} + 4k\pi + \pi$

Hence $I(k) = \left(\frac{8\pi^3}{3} \right) k^2 + (4\pi)k + \pi$

This is a quadratic in k and its minimum value

occurs when $k = \frac{-4\pi \cdot 3}{16\pi^3} = -\frac{3}{4\pi^2}$

$\therefore f(x) = \left(-\frac{3}{4\pi^2} \right) x ; f(-4\pi^2) = 3$

5. Ans (3.50)

Obviously, f is even, so we find those a , b and c , which make f to be continuous for $x \in [0, 2]$.

Now, $f(x) = \begin{cases} ax^2 + b & , 0 \leq x < 1 \\ 1 & , x = 1 \\ \frac{c}{x} & , 1 < x \leq 2 \end{cases}$

So, continuity of f at $x = 1$

$\Rightarrow a + b = 1 = c \dots(1)$

Also, $f'(x) = \begin{cases} 2ax & , 0 < x < 1 \\ -\frac{c}{x^2} & , 1 < x < 2 \end{cases}$

So, differentiability of f at $x = 1$

$\Rightarrow 2a = -c \dots(2)$

$\therefore$ On solving (1) and (2),

we get $a = \frac{-1}{2}, b = \frac{3}{2}, c = 1$

Since, f is even, so $f(-2) = f(2)$ is already true.

Thus, Rolle's theorem is applicable to f in $[-2, 2]$,

if $a = \frac{-1}{2}, b = \frac{3}{2}, c = 1$

Hence, $(b^2 + c^2 + a^2) = 3.50$

6. Ans (1.50)

$\Rightarrow \int \{ (x^6 + 6x^5) + (x^5 + 5x^4) + (x^4 + 4x^3) + (x^3 + 3x^2) + (x^2 + 2x) + (x + 1) \} e^x dx$

$\Rightarrow (x^6 + x^5 + x^4 + x^3 + x^2 + x)e^x + C$

$\therefore \int e^x (f(x) + f'(x)) dx = e^x f(x) + C$

$= \sum_{k=1}^6 x^k e^x + C$

$\Rightarrow \alpha = 6, \beta = 1$